arising below the point of injury; in fact, it is manifest that the stupendous powers of motion and sensation find their instrumentalities in these little insignificant-looking nerve-filaments; sever, wound, or injure them, and paralysis of motion, or sensation, is just as inevitably the result as if the columns of the medulla oblongata were cut away as before described.

The nerves arise, as we have said, in pairs; a nerve of motion and sensation in one-half of the column, and corresponding roots springing from the other half,—the sensory nerve-roots arising from the posterior division, the motor nerve-roots from the anterior.

These pairs arise at first singly, but they soon after combine into a ganglionic knot, again to divide and subdivide and ramify into countless fibrous lengths, supplying every part of the body, reaching to the ultimate points of the spine, the interior of the bones, muscles, membranes, tissues, and blood-vessels: in fact, it is impossible to conceive any portion of the organism which is not supplied with some ramification, great or small, from the main trunks of the cerebro-spinal system.

It must be noted that nervous action is invariably dual or reflex: for example, the cerebrum wills to move a joint or contract a muscle; the thought is immediately telegraphed through the medulla oblongata to the desired point, and whatever sensation is produced by that motion is instantaneously telegraphed back again to the brain by the corresponding nerve of *the pair*. Of course there are a vast number of motions going on in the system which proceed wholly independent of volition. Many of these arise